

Scale-Coherence Rigidity and the Orbitwise Gap in Collatz Dynamics

Paper D: From Population Mixing to Individual Orbits

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Abstract

Papers B and C established exact population-level mixing for the Collatz dynamics modulo 2^K : the distribution of $T^t(a) \bmod 2^K$ converges to uniform in $K - 1$ steps (TV sense), and a single high-bit flip already creates a spectral gap $\delta_{K,1} \approx 0.29$ independent of K . The present paper addresses the fundamental gap between these *population* statements and the *individual orbit* statement required by the Collatz conjecture.

We identify three layers of structure. First, for any fixed orbit $\{n_t\}$, the effective high-bit parameter $m(t) = \lfloor \log_2 n_t \rfloor - K$ grows with n_t , making the K -level dynamics increasingly close to the true 2-adic Haar kernel—which, paradoxically, has *exponentially small* spectral gap. Second, Weyl equidistribution implies that $n_t \bmod 2^K$ visits each residue with limiting frequency $1/n_K$, including residue 1; but visiting $n_t \equiv 1 \pmod{2^K}$ does not imply $n_t = 1$. Third, the Collatz conjecture is equivalent to the *simultaneous scale condition*: $n_t \equiv 1 \pmod{2^K}$ for *all* K simultaneously forces $n_t = 1$.

We formulate this as the **Simultaneous Scale Coherence Problem** and show that its resolution—which would imply the Collatz conjecture—requires exactly the multi-scale information rigidity identified in Papers B and C. We prove a conditional theorem: if the Open Gap Problem of Paper C holds ($\inf_K \delta_{K,1} > 0$), then the arithmetic memory requirements for a divergent orbit become increasingly contradictory across scales.

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1 Introduction

1.1 The series so far

This is the fourth paper in a series on the arithmetic mixing geometry of Collatz dynamics.

Paper A [1] established the renewal operator structure: the Syracuse map has a natural renewal decomposition, and the corresponding transfer operator has exact spectral properties.

Paper B [2] proved the exact TV mixing formula: for the Markov kernel P_K on odd residues modulo 2^K induced by the 2-adic Haar measure, the support of $P_K^t(a, \cdot)$ forms an exact arithmetic progression, and

$$\text{TV}(P_K^t(a, \cdot), \pi_K) = \max(0, 1 - 2^{S_t(a) - (K-1)}),$$

yielding mixing time $T_{\text{mix}}(K) = K - 1$.

Paper C [3] introduced the one-bit high-bit kernel $P_{K,1}$ and observed the spectral jump $\delta_{K,0} = 0 \rightarrow \delta_{K,1} \approx 0.29$, formulating the Open Gap Problem: $\inf_K \delta_{K,1} > 0$.

The present paper asks: *what do these population-level mixing results imply for individual Collatz orbits?*

1.2 The fundamental gap

Population-level mixing (Papers B and C) makes statements about distributions:

If a is drawn uniformly at random from Ω_K , then $T^t(a) \bmod 2^K$ is close to uniform after $t = K - 1$ steps.

The Collatz conjecture requires an individual-level statement:

For every fixed positive integer n_0 , the orbit $n_0, T(n_0), T^2(n_0), \dots$ eventually reaches 1.

The gap between these levels is *genuine*: a single deterministic orbit is not a random variable, and population mixing does not directly imply orbitwise convergence. Closing this gap is the central challenge of this paper.

1.3 Main contributions

1. **Effective high-bit parameter $m(t)$:** We introduce $m(t) = \lfloor \log_2 n_t \rfloor - K$ as the effective number of high bits active at time t , and show that for divergent orbits, $\limsup_t m(t) = \infty$ while $m(t) \rightarrow \infty$ drives the K-level dynamics toward the true Haar kernel—which has *exponentially small* spectral gap (Section 2).
2. **The self-mixing paradox:** Divergent orbits are in a sense *poorly* mixing at each scale K , precisely because their large values push $m(t)$ into the regime where the Haar spectral gap is small. This reverses naive intuition (Section 3).
3. **Weyl equidistribution and its limits:** We connect to the known Weyl equidistribution result for Collatz orbits, and identify precisely why it does not suffice for the Collatz conjecture (Section 4).
4. **The Simultaneous Scale Coherence Problem:** We show the Collatz conjecture is equivalent to a multi-scale simultaneous condition, and formulate this as a precise open problem (Section 5).
5. **Conditional theorem:** If the Open Gap Problem of Paper C holds, then divergent orbits require an arithmetically impossible simultaneous coherence structure (Section 6).

2 The Effective High-Bit Parameter

Definition 2.1 (Effective high-bit parameter). For an odd positive integer n_t and a scale $K \geq 1$, the *effective high-bit parameter at time t* is

$$m(t) = m(t; K) := \max(0, \lfloor \log_2 n_t \rfloor - K).$$

When $n_t \leq 2^K$, we have $m(t) = 0$ and the K-level dynamics is exactly deterministic: $n_{t+1} \bmod 2^K = F_K(n_t \bmod 2^K)$. When $n_t > 2^K$, the high bits of n_t inject $m(t)$ bits of additional information into the K-level transition, and the K-level dynamics is governed by the kernel $P_{K, m(t)}$ of Paper C.

Proposition 2.2 (High-bit inflow formula). *For any odd n_t with $m(t) \geq 1$:*

$$n_{t+1} \bmod 2^K = F_K(n_t \bmod 2^K) + 3u_t \cdot 2^{K-v_t} \bmod 2^K,$$

where $v_t = v_2(3n_t + 1)$ and $u_t = \lfloor n_t / 2^K \rfloor \bmod 2^{m(t)}$ encodes the active high bits.

Proof. Write $n_t = (n_t \bmod 2^K) + u_t \cdot 2^K$ with $0 \leq u_t < 2^{m(t)}$. By Lemma V-stability (Paper C, Lemma 2.3), $v_2(3n_t + 1) = v_t = v_2(3(n_t \bmod 2^K) + 1)$ when $v_t < K$. The formula follows by expanding $T(n_t) = (3n_t + 1)/2^{v_t}$ and reducing modulo 2^K . $\square$

2.1 Behavior for divergent orbits

Proposition 2.3 (Unbounded $m(t)$ for divergent orbits). *If $\{n_t\}$ is a divergent Collatz orbit ($n_t \rightarrow \infty$), then for every $K \geq 1$:*

$$\limsup_{t \rightarrow \infty} m(t; K) = \infty.$$

Proof. Since $n_t \rightarrow \infty$, for any M there exist infinitely many t with $n_t > 2^{K+M}$, so $m(t; K) > M$. $\square$

The following table illustrates $m(t)$ for the orbit of $n_0 = 27$ at scale $K = 6$:

t	n_t	$n_t \bmod 2^6$	$m(t)$	$\delta_{K,m(t)}$ (approx.)
0	27	27	0	0
4	71	7	0	0
6	161	33	1	0.313
9	137	9	1	0.313
14	263	7	2	0.428
16	593	17	3	0.571

As n_t grows, $m(t)$ increases and the effective spectral gap $\delta_{K,m(t)}$ increases—until the orbit falls back to small values.

3 The Self-Mixing Paradox

3.1 Large $m(t)$ does not imply fast mixing

One might expect that a large $m(t)$ —many high bits—implies fast K -level mixing. Paper C shows the opposite.

Proposition 3.1 (Self-mixing paradox). *As $m(t) \rightarrow \infty$, the effective kernel $P_{K,m(t)}$ approaches the true 2-adic Haar kernel P_K^{Haar} , which has spectral gap $\delta_K^{\text{Haar}} \approx 2^{-(K-1)} \rightarrow 0$ exponentially in K .*

In other words: *divergent orbits become progressively harder to mix spectrally at each scale K , because their large values push the K -level dynamics into the Haar regime with exponentially small spectral gap.*

This contrasts with the one-bit regime ($m = 1$), where the spectral gap is $\delta_{K,1} \approx 0.29$ uniformly in K . The phase diagram (Paper C, Section 3) makes this precise:

$m(t)$	Kernel regime	Spectral gap $\delta_{K,m}$
0	Deterministic F_K	0
1	One-bit flip	≈ 0.29
K	Arithmetic average	≈ 0.74
∞ (arith.)	Saturated average	≈ 0.74
∞ (true Haar)	2-adic Haar	$\approx 2^{-(K-1)} \rightarrow 0$

The regime relevant for divergent orbits ($m(t) \rightarrow \infty$) is the bottom row: worst for spectral mixing.

3.2 Interpretation

Paper C showed: *structured finite inflow beats full Haar randomization*. The present paper refines this: divergent orbits are in the Haar regime, which is the least efficient for spectral decorrelation. This means divergent orbits *self-inhibit* their own mixing—a structural constraint that will be exploited in the conditional theorem (Section 6).

4 Weyl Equidistribution and Its Limits

4.1 What equidistribution gives

A classical result (Terras [6], Lagarias [4]) states that Collatz orbits are *equidistributed* modulo 2^K :

Theorem 4.1 (Weyl equidistribution, after Terras–Lagarias). *For any positive odd integer n_0 and any odd residue $r \in \Omega_K$:*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \#\{t \leq T : n_t \equiv r \pmod{2^K}\} = \frac{1}{n_K},$$

assuming the orbit $\{n_t\}$ is infinite (i.e., assuming n_0 is divergent or the orbit is long before reaching 1).

In particular, for $r = 1$: every infinite orbit visits $n_t \equiv 1 \pmod{2^K}$ with limiting frequency $1/n_K$.

4.2 Why this does not suffice

The condition $n_t \equiv 1 \pmod{2^K}$ means $n_t \in \{1, 1 + 2^K, 1 + 2 \cdot 2^K, \dots\}$. For a divergent orbit, only values $n_t = 1 + j \cdot 2^K$ with $j \geq 1$ are visited (not $n_t = 1$ itself). The equidistribution theorem does not distinguish between $n_t = 1$ and $n_t = 1 + j \cdot 2^K$ for $j \geq 1$.

Remark 4.2 (The equidistribution gap). Theorem 4.1 gives frequency $1/n_K$ of visiting $n_t \equiv 1 \pmod{2^K}$, but does *not* imply that $n_t = 1$ is ever reached. A divergent orbit can satisfy the equidistribution theorem at every scale K while never actually reaching 1.

This gap is not a deficiency of the argument; it is intrinsic to frequency-based methods applied to the integer 1.

5 The Simultaneous Scale Coherence Problem

5.1 2-adic characterization of convergence

The key observation is algebraic: the positive integer 1 is the *unique* positive integer satisfying

$$n \equiv 1 \pmod{2^K} \quad \text{for all } K \geq 1.$$

This is because the 2-adic integers $\mathbb{Z}_2$ embed $\mathbb{Z}_{>0}$ via the map $n \mapsto (n \bmod 2^K)_{K \geq 1}$, and the 2-adic representation of 1 is $(1, 1, 1, \dots)$. Any other positive integer $m \geq 3$ fails $m \equiv 1 \pmod{2^K}$ for $K = \lfloor \log_2(m-1) \rfloor + 1$.

Proposition 5.1 (2-adic criterion for convergence). *A Collatz orbit $\{n_t\}$ converges to 1 if and only if:*

$$\exists t_0 : \forall K \geq 1, \quad n_{t_0} \equiv 1 \pmod{2^K}.$$

Equivalently: $\exists t_0 : n_{t_0} = 1$.

5.2 The simultaneous scale condition

Definition 5.2 (Scale-coherence at 1). An orbit $\{n_t\}$ is (K, R) -coherent at 1 if there exists $t_0 \leq R$ such that $n_{t_0} \equiv 1 \pmod{2^K}$.

It is (∞, R) -coherent at 1 if there exists $t_0 \leq R$ such that $n_{t_0} \equiv 1 \pmod{2^K}$ for **all** $K \geq 1$ simultaneously.

By Proposition 5.1, (∞, R) -coherence at 1 is equivalent to $n_{t_0} = 1$ for some $t_0 \leq R$.

Open Problem 5.3 (Simultaneous Scale Coherence Problem). Let $\{n_t\}$ be any Collatz orbit. Prove:

$$\forall K \geq 1, \exists t_0 : n_{t_0} \equiv 1 \pmod{2^K} \implies \exists t_0 : n_{t_0} = 1.$$

Remark 5.4. The left-hand side of Open Problem 5.3 is a consequence of Theorem 4.1: by equidistribution, every infinite orbit visits $n_t \equiv 1 \pmod{2^K}$ for each fixed K . The challenge is the *simultaneity*: finding a single t_0 that works for all K at once.

Remark 5.5 (Equivalence to Collatz). Open Problem 5.3 is equivalent to the Collatz conjecture. The implication “ $\exists t_0 : n_{t_0} = 1 \Rightarrow$ each- K condition” is trivial. The converse is what needs proof.

The following table illustrates how the constraint tightens with K :

K	Mod 2^K constraint	Minimum $n > 1$ satisfying it
1	$n \equiv 1 \pmod{2}$	3
4	$n \equiv 1 \pmod{16}$	17
8	$n \equiv 1 \pmod{256}$	257
12	$n \equiv 1 \pmod{4096}$	4097
K	$n \equiv 1 \pmod{2^K}$	$1 + 2^K$

A divergent orbit satisfying the constraint at every K simultaneously would need $n_{t_0} > 2^K$ for all K —impossible for any finite n_{t_0} .

6 Conditional Theorem

We now state the main conditional result, connecting the Open Gap Problem (Paper C) to the Simultaneous Scale Coherence Problem.

6.1 Setup

Let $\{n_t\}$ be a Collatz orbit. At time t and scale K , the effective kernel is $P_{K,m(t)}$ with spectral gap $\delta_{K,m(t)}$. Define the *cumulative spectral mixing* up to time T at scale K as:

$$\Delta_K(T) := \sum_{t=1}^T \delta_{K,m(t)}.$$

Theorem 6.1 (Conditional Scale-Coherence Rigidity). *Assume the Open Gap Problem of Paper C holds: $\inf_K \delta_{K,1} \geq c > 0$.*

Let $\{n_t\}$ be a Collatz orbit with $\liminf_{t \rightarrow \infty} n_t > 2^K$. Then:

$$\Delta_K(T) \geq c \cdot \#\{t \leq T : n_t > 2^K\} \rightarrow \infty \quad \text{as } T \rightarrow \infty.$$

In particular, the K -level spectral mixing accumulated by the orbit diverges, implying that the K -level distribution of $n_t \bmod 2^K$ becomes asymptotically indistinguishable from the stationary distribution $\pi_{K,m}$.

Proof. For each t with $n_t > 2^K$, we have $m(t; K) \geq 1$, so $\delta_{K,m(t)} \geq \delta_{K,1} \geq c$. Summing over all such t gives $\Delta_K(T) \geq c \cdot \#\{t \leq T : n_t > 2^K\}$. If $\liminf_{t \rightarrow \infty} n_t > 2^K$, then almost all t satisfy $n_t > 2^K$, so the count grows as T . $\square$

6.2 Interpretation and the remaining gap

Theorem 6.1 establishes that divergent orbits accumulate diverging spectral mixing at each fixed scale K . This means the K -level time-averaged behavior approaches stationarity.

However—and this is the honest statement of the remaining gap—accumulated spectral mixing in the *population* (or time-averaged) sense does not directly force the *individual* orbit to hit $n_t = 1$. The theorem shows increasing decorrelation in the time average, but the orbit remains a single deterministic trajectory.

Remark 6.2 (The remaining gap). Closing the gap between Theorem 6.1 and a proof of the Collatz conjecture requires showing that diverging cumulative spectral mixing at *all* scales simultaneously forces a single time t_0 where $n_{t_0} = 1$. This is precisely the Simultaneous Scale Coherence Problem (Open Problem 5.3).

The contribution of this paper is to make the gap explicit and to show that it is not a vague philosophical distinction but a precise mathematical problem: the transition from “each- K equidistribution” to “all- K simultaneously”.

7 The Structure of Papers B, C, D

We now articulate the logical structure of the series, showing how Papers B, C, D form a coherent program.

Paper	Core result	Status
B	Exact TV mixing: $\text{TV}(P_K^t, \pi_K) = \max(0, 1 - 2^{S_t - (K-1)})$	Proved
C	One-bit spectral jump: $\delta_{K,0} = 0 \rightarrow \delta_{K,1} \approx 0.29$	Numerical; Open Gap Problem
D	Simultaneous scale coherence: divergence requires impossible memory	Open; conditional on C

The logical dependencies are:

$$\underbrace{\text{Paper B (ETV)}}_{\text{proved}} \longrightarrow \underbrace{\text{Theorem D (conditional)}}_{\text{conditional on C}} \longleftarrow \underbrace{\text{Paper C (Open Gap)}}_{\text{open}}.$$

If Paper C’s Open Gap Problem is resolved, Theorem 6.1 gives diverging spectral mixing for all orbits with $\liminf n_t > 2^K$. The remaining bridge—Simultaneous Scale Coherence (Open Problem 5.3)—is equivalent to the Collatz conjecture itself.

8 Summary of Results

Result	Status	Section
Effective high-bit parameter $m(t)$	Definition	§2
Proposition: $\limsup m(t) = \infty$ for divergent orbits	Proved	§2
Self-mixing paradox (large $m \Rightarrow$ small gap)	Proved	§3
Weyl equidistribution and its limits	Known/Explained	§4
2-adic criterion for convergence	Proved	§5
Simultaneous Scale Coherence Problem	Open (= Collatz)	§5
Conditional Theorem (Open Gap $\Rightarrow$ diverging mixing)	Conditional	§6
The remaining gap (population $\nRightarrow$ individual)	Identified	§6

References

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